# Akshay Bagde

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## EDUCATION

| Institution | Degree / Board | Passing Year | Score / CGPA |
|---|---|---|---|
| Indian Institute of Technology Hyderabad | B.Tech | 2027 | 7.48 |
| Kendriya Vidyalaya No. 1, Chhindwara | Class XII (CBSE) | 2020 | 78.20% |
| Kendriya Vidyalaya No. 1, Chhindwara | Class X (CBSE) | 2018 | 81.60% |

## TECHNICAL SKILLS

**Programming Languages**: C++, Python, C, SQL

**Tools & Platforms**: Linux, Git, GitHub, CMake, Bash, ChampSim

**Programming Concepts**: Object-Oriented Programming, STL

**Libraries & Frameworks**: PyTorch, OpenCV, NumPy, Pandas, scikit-learn

**CS Fundamentals:** Operating system, DBMS, Computer Networks, Computer Architecture

## M.TECH THESIS

**Cache Partitioning and Replacement Policy Co-Design for Multicore Systems**

**Supervisor:** Dr. Shirshendu Das

**Domain:** Computer Architecture, Cache Systems, Multicore Processors, Performance Optimization

- Implemented and evaluated LRU, SRRIP, Hawkeye, Mockingjay and CHROME replacement policies with UCP-based cache partitioning in ChampSim.

- Currently investigating the co-design of dynamic way partitioning with Mockingjay and CHROME for shared multicore LLCs.

- Analyzed partition-restricted victim selection and partition-aware bypassing, cross-core interference and stale policy state after repartitioning.

- Conducted trace-driven multicore simulations using SPEC CPU traces to compare LLC misses and IPC across replacement-only and partition-aware configurations.

**Tools & Technologies:** C++, ChampSim, Linux, Bash, Git, Python

## PROJECTS

**MiniDB - Disk-Backed Relational Storage Engine**

- Built a C++ disk-backed relational engine supporting CRUD operations across tables with persistent binary storage.

- Implemented a B+ Tree primary-key index mapping record IDs to disk offsets for O(log N) point lookups.

- Added an LRU row cache, multi-table catalog management, soft deletion and automatic index rebuilding after restarts.

**Secure Peer-to-Peer Payment Platform | Network Security**

- Co-developed a peer-to-peer payment platform supporting secure authentication, profile management, money transfers and transaction history.

- Implemented prepared SQL statements, bcrypt password hashing, CSRF protection, secure session handling and server-side input validation.

- Built atomic MySQL transactions with balance validation and rollback on failure, and containerized the application

using Docker Compose.

**Binary Image Classification on Synthetic 3D Scenes**

- Built a dual-stream ResNet-34 pipeline combining RGB features with geometry-based Canny and Sobel representations for binary scene classification.

- Implemented MixUp, Stochastic Weight Averaging, 4-view test-time augmentation and F1-based threshold tuning to improve model generalization.

- Integrated probability-based ensembling and semi-supervised pseudo-labeling, achieving 81.35% accuracy.

## RELEVANT COURSES

- **Theory:** Advanced Data Structures and Algorithms

**- Systems:** Advanced Computer Networks, Network Security

**- AI & Data Science:** Foundations of Machine Learning, Deep Learning, Visual Computing , Representation Learning, Fraud Analytics

## SCHOLASTIC ACHIEVEMENTS

- Solved **700+ coding problems** on LeetCode.
- Secured **6th rank** in the **IIT Hyderabad Deep Learning Hackathon 2026**.
- Secured **13th rank** in the **KLA AI Hack x IIT Hyderabad 2026**.

## POSITIONS OF RESPONSIBILITY

**Teaching Assistant**

Courses: Introduction to Programming (C), Approximation Algorithms, Advanced Data Structures and Algorithms

- Evaluated C programming and algorithmic assignments with emphasis on correctness, data structures, complexity analysis and problem-solving approach.

- Reviewed and graded solutions involving advanced algorithms and optimization techniques, while assisting students with debugging and conceptual clarification.

## EXTRACURRICULAR ACTIVITIES

**- Cricket:** Two-time KVS National-level player; Runner-up in the Inter College League (ICL).